

# When Clustered Extremes Do Not Become Early Warnings: A Negative-Result Dynamical-EVT Study for Public Industrial Monitoring Data

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## Abstract

This paper evaluates whether clustered extreme scores from a dynamical-EVT-inspired workflow become useful maintenance alarms on public industrial monitoring data. The study combines high-replication Monte Carlo checks across fourteen process families, held-out compressor event evaluation, target-region variants, matched negative controls, and non-event stress tests. The primary empirical evidence contains two independent held-out compressor failures: one in MetroPT and one in MetroPT2. The registered recurrence score detected both, but with 2631 and 852 alarm episodes, false-alarm burdens of 105.34 and 51.47 alarms per operating day, and event precision below 0.12%. The best simple engineering threshold also detected both events, and whether it reduces alarm burden depends on which burden is measured: it raises fewer alarm episodes on MetroPT (1551 against 2631) while holding the system under warning for 75.1% of the test period against the registered score's 0.12%. Episode counts and exposure disagree in direction here, so neither alone establishes an operational advantage. The strongest target-region result was heterogeneous: the failure-prototype region missed the held-out MetroPT event entirely, but detected the MetroPT2 event with 157 alarms, 9.41 false alarms per day, and precision 0.00637. Matched controls show that this MetroPT2 burden is lower than rare-region and phase-randomised controls, but not uniformly better than occupancy-matched random regions. The narrow conclusion is therefore negative and conditional: recurrence-based EVT diagnostics can expose finite-sample estimator instability, target-region heterogeneity, and alarm-conversion failure, but these public datasets do not establish operationally reliable early warning.

## 1 Introduction

Industrial maintenance alarms are useful only when they map to the decision unit that operators and engineers actually face. A pointwise anomaly score may flag thousands of telemetry samples around one physical failure, while the operator sees one alarm burden, one repair decision, and one opportunity for useful warning. The same difficulty appears outside compressor telemetry: a fleet repair record, a laboratory hydraulic cycle, and a semiconductor yield label are not exchangeable evidence units. Any empirical claim must therefore separate samples, exceedances, clusters, alarm episodes, failures, vehicles, cycles, and production examples.

Dynamical extreme-value theory offers a tempting route from recurrent machine states to early-warning logic. If an observable is large when the trajectory returns near a target set, then recurrence and periodicity can cluster extremes [4, 5]. In industrial monitoring, however, clustered extremes can also arise from operating regimes, preprocessing, smoothing, missingness, persistent shifts, or alarm-conversion rules. Noise can further weaken the connection between deterministic recurrence and observed extremes [2]. The scientific question is therefore not whether an extremal-index diagnostic can be computed. The question is when such a diagnostic translates into useful episode-level warning.

This paper is framed as a negative-result study. It asks four research questions. First, does threshold and regime conditioning stabilise extremal-index diagnostics relative to global thresholding? Second, does declustering reduce duplicate alarm episodes without destroying event recall or warning lead time? Third, does the registered robust extreme-score diagnostic add operational value after episode conversion? Fourth, which mechanisms explain empirical failure: target-region misspecification, regime mixture, too few independent failures, non-extreme fault signatures, label ambiguity, calibration shift, or alarm conversion?

The evaluated workflow combines causal state construction, training-only target-region construction, regime-conditioned thresholds, extremal-index diagnostics, multivariate lag patterns, alarm merging, event matching, baseline provenance, and dataset-specific condition diagnostics. The interpretation is deliberately conservative so that negative and non-estimable outcomes remain visible.

The contribution is narrow. The study shows that dynamical-EVT-inspired diagnostics can reveal dependence, threshold sensitivity, duplicate-alarm behavior, and calibration limits, but the current public-data evidence does not show reliable early warning. MetroPT and MetroPT2 are the primary event-level case studies; SCANIA Component X is an external vehicle-level stress test; Hydraulic Systems and SECOM are supplementary condition-monitoring demonstrations. This separation is essential because the datasets support different estimands. The paper therefore reports what failed, where the evidence is insufficient, and which conditions would be required before making broader predictive-maintenance claims.

## 2 Related Work

Predictive-maintenance models often begin with supervised classification, reconstruction error, distance-based anomaly detection, or streaming thresholds. These methods are useful engineering baselines, but their common output is a timestamped score rather than an episode-level maintenance decision. EVT-based streaming detectors such as SPOT use peaks-over-threshold modelling to adapt anomaly thresholds in streams [10]; they address threshold calibration, but they do not by themselves resolve operating-regime mixture, duplicate alarms per failure, or the distinction between clustered observations and independent maintenance events.

Classical declustering and extremal-index estimation address dependence among extremes. Ferro and Segers proposed an inference scheme for clusters of extreme values using inter-exceedance times and the extremal index [3]. In dynamical systems, hitting-time statistics connect returns to target neighborhoods with extreme-value laws [4]. Periodic points can generate an extremal index smaller than one, giving a mathematical route from recurrence to clustered extremes [5]. The finite-sample industrial setting is less clean: noise may make deterministic recurrence appear closer to independence, and observed clustering can also arise from smoothing, missingness, persistence, or regime mixture [2]. Recent multivariate work extends dynamical EVT to cross-sectional dependence among vector-valued observables [1], which motivates the lag-pattern diagnostics used here.

The industrial datasets used by the repository are public condition-monitoring resources with different independence units and labels. MetroPT is an air-production-unit telemetry dataset for predictive maintenance [12]. MetroPT2 is a replication-style benchmark released through Zenodo [13]. SCANIA Component X is a fleet-level multivariate time-series dataset with operational histories and repair records for an anonymized component [7]. UCI Hydraulic Systems provides laboratory hydraulic test-rig cycles with component-condition targets [6]. UCI SECOM provides semiconductor process measurements with pass/fail yield labels [8]. These datasets broaden the empirical base, but they do not remove the need for estimand-specific accounting: one long compressor fault interval, one truck repair history, one hydraulic load cycle, and one wafer-level yield example are not interchangeable statistical units. The methodological gap addressed here is the absence of a reproducible layer that connects dynamical extreme

diagnostics to alarm episodes, tabular condition diagnostics, sensitivity analyses, and claim constraints.

### 3 Method

Let  $Z_t \in \mathbb{R}^p$  be the causal state vector at sample  $t$ . For compressor telemetry,  $Z_t$  contains the registered raw pressure, temperature, current, and flow channels used by the real-data diagnostic. For tabular condition-monitoring adapters,  $Z_t$  contains the registered feature vector for the cycle, vehicle, or production example. Scaling is fitted on the training partition only: each feature is centered by the training median and divided by the training interquartile range with a positive floor for zero-spread features.

Operating regimes  $R_t$ , when used in the sensitivity assets, are inferred without test labels. Regimes are conditioning variables for threshold estimation and are not interpreted as mechanical causes. The registered real-data matrix reported in Tables 7 and 11 uses a conservative train-split robust score rather than the full dangerous-region construction.

The implementation also provides dangerous-region scoring for separate diagnostics. The failure-proximal prototype variant uses only training failure intervals and records the source interval for every prototype. The unsupervised rare-state variant ignores failure labels and fits a high-distance rare-state set from training data only. These variants are available for diagnostic analysis and ablation coverage, but the registered real-data matrix in this manuscript is the conservative high-quantile score; it should not be read as evidence that both dangerous-region variants have already been optimized on every dataset.

For a target region  $A$ , the scalar observable is

$$X_t = -\log\{d(Z_t, A) + \epsilon\},$$

where  $d$  is the robust Euclidean distance in scaled coordinates and  $\epsilon > 0$  prevents singularities. Larger  $X_t$  means closer proximity to the training-defined region. This is an anomaly or recurrence score, not a calibrated probability of failure.

For each regime  $r$ , a threshold  $u_r(q)$  is estimated from the declared training or calibration partition at quantile  $q$ . Exceedances are  $I_t(q, r) = \mathbf{1}\{X_t > u_r(q), R_t = r\}$ . Runs declustering merges exceedances separated by at most  $m$  samples. The runs estimator, Ferro–Segers inter-exceedance estimator, and K-gaps-style declustering diagnostic summarize clustering under the frozen  $q$  and  $m$  settings [3, 11]. The inverse extremal index is discussed only as an asymptotic cluster-size diagnostic under the assumptions of the estimator.

Pointwise exceedances are converted into alarm episodes by the declared merge gap. A test alarm is a true event alarm only if the episode intersects the predeclared warning window for a held-out failure under the frozen matching tolerance. All event recall, event precision, false-alarm burden, duplicate-alarm count, lead time, time under warning, and utility metrics are computed after this episode conversion when event labels exist. This prevents timestamp counts from being treated as independent maintenance evidence.

The score-stratification target for horizon  $h$  is

$$Y_t^{(h)} = \mathbf{1}\{\text{a labelled event begins within } h \text{ after } t\}.$$

In the current evidence, model outputs are anomaly ranks or recurrence scores. Probability-calibration language is therefore restricted: Brier-style summaries are reported only as diagnostics of score stratification unless a separate calibration partition contains enough positive independent units for an explicit probability map.

The leakage contract is: training features cannot contain held-out failure samples; scaling, regimes, target regions, thresholds, lag selection, merge-gap selection, and calibration cannot use test events; SCANIA vehicles cannot appear in multiple partitions; and feature windows

cannot cross maintenance-reset boundaries. Table 1 gives the terminal audit result for each registered check. Under the current registered diagnostic the algorithm is:

1. Split entities or intervals into training, validation, calibration, and test partitions.
2. Fit scaling and robust score thresholds on allowed partitions only.
3. Select  $q, m$ , and merge gap on validation/calibration partitions only where used.
4. Freeze all parameters and score the held-out test partition causally.
5. Extract exceedance clusters, merge alarm episodes, and match them to labelled events.
6. Report event metrics, uncertainty by independent unit, and non-estimable cases.

Table 1: Terminal leakage-audit results for the registered workflow.

| Check                                                      | Result | Evidence                               |
|------------------------------------------------------------|--------|----------------------------------------|
| Held-out failures excluded from feature construction       | passed | causal trailing windows                |
| Held-out failures excluded from scaling                    | passed | train-split robust scaling             |
| Held-out failures excluded from regime inference           | passed | unsupervised labels excluded           |
| Held-out failures excluded from target-region construction | passed | train-only prototypes                  |
| Test labels excluded from threshold selection              | passed | train/calibration quantiles            |
| Test labels excluded from lag selection                    | passed | registered lag grids                   |
| Test labels excluded from merge-gap selection              | passed | frozen evaluation protocol             |
| Test labels excluded from calibration                      | passed | validation-only score mapping          |
| Maintenance boundaries respected                           | passed | boundary fields in protocol            |
| SCANIA vehicles disjoint across partitions                 | passed | preparation raises on overlap          |
| No future information in causal windows                    | passed | current and trailing observations only |

The practical complexity for scoring is  $O(npK)$ , where  $n$  is the number of samples or entities,  $p$  is the state dimension, and  $K$  is the number of stored prototypes or rare-state summaries. Thresholding and runs declustering are linear in  $n$ . The streaming-compatible parts are causal feature construction, scaling with frozen parameters, scoring, thresholding, and alarm merging. Prototype construction, model selection, and final event matching are offline evaluation steps.

## 4 Simulation Study

The simulation layer is designed to falsify overconfident interpretations before industrial data are used. The registered systems include independent extremes, periodic dynamics, noisy periodic dynamics, regime mixtures, missingness perturbations, smoothing and downsampling perturbations, and degradation-like cyclic trajectories. Each simulation cell records its configuration, random seed, threshold grid, run-length grid, estimator status, and raw outputs sufficient to regenerate the paper assets.

## 4.1 Estimators under comparison

Six estimators of the extremal index are compared: runs declustering, the Ferro–Segers intervals estimator, K-gaps, the reciprocal mean block-cluster size, the disjoint-blocks estimator, and a no-declustering reference fixed at  $\theta = 1$ . The last is not a competitive estimator but a control: any procedure that cannot separate a clustered process from an independent one should be indistinguishable from it. The runs and K-gaps estimators are tuned by the declustering run length; the block-based estimators are tuned by the block size, which is set to  $\lceil \sqrt{n} \rceil$  so that the tuning parameter is declared rather than fitted.

The reciprocal mean cluster size is computed over disjoint blocks rather than over runs-declustered clusters. This is deliberate. When clusters are taken from runs declustering they partition the exceedances exactly, so the reciprocal mean cluster size reduces algebraically to the runs estimator and carries no independent information; reporting the two as separate estimators would present one quantity twice.

## 4.2 Reference extremal indices

Bias, root mean squared error and coverage all require a reference value of  $\theta$ . Closed-form references exist only for the independent systems and for the noiseless logistic map at its periodic target, where  $\theta = 1/2$ . Every noisy configuration lies outside those cases, so references there are established numerically from a long realisation of  $4 \times 10^5$  samples.

The numerical construction is deliberately conservative, because a reference obtained from one declustering rule is not independent of the estimators being judged: scoring against a runs-based reference would favour the runs estimator by construction. Two structurally different long-run estimates are therefore computed, one by runs declustering at a short run length and one by disjoint blocks at a block scaled to hold roughly one expected exceedance. A numerical reference is accepted only where the two agree within 0.05; otherwise the configuration is recorded as one where  $\theta$  could not be established, and it contributes no bias, RMSE or coverage evidence. Both long-run settings are calibrated against the two systems whose value theory already fixes, recovering 0.495 against a theoretical 0.5 and 0.961 against a theoretical 1.0.

Theoretical and numerical references are never pooled within a reported row, and every table states which kind it used.

## 4.3 Interval estimation

None of the six estimators has a usable closed-form interval in this setting, so intervals are obtained by a circular block bootstrap over the exceedance indicator series. Resampling whole blocks preserves the short-range dependence that the extremal index measures; an independent bootstrap would destroy exactly the structure under study and return intervals centred on one. Resamples that fail to yield an admissible estimate are counted rather than discarded, so the reported resample success rate is honest.

## 4.4 What this grid can and cannot establish

The simulation evidence is not uniform across the clustering regimes, and the reported scope is limited accordingly. The high-replication grid runs at a single noise level, and at that noise level the numerically established references sit close to independence: of its nine referenced process families, eight have  $\theta \geq 0.9$ . Its 500 replications per cell therefore buy precision about *near-independent* processes. Clustered evidence does exist — the cyclic degradation family is established between  $\theta = 0.049$  and  $\theta = 0.175$ , and the smoothing family at  $\theta = 0.272$  — but the strongly clustered cells carry far fewer replications, so their coverage estimates are correspondingly imprecise.

Accordingly, this study reports the high-replication grid as evidence about near-independent processes, and treats the clustered cells as supporting rather than primary evidence. Estimator failures, non-estimable quantities and unestablished references are preserved in the artifacts as part of that scope statement rather than removed from it.

## 5 Datasets

The datasets are assigned roles before interpretation. MetroPT and MetroPT2 are the primary event-level case studies because they contain ordered compressor telemetry and labelled failure periods [12, 13]. They support early-warning, alarm-episode, recurrence, and per-failure questions, but their independent event count is small. SCANIA Component X is an external vehicle-level stress test with irregular fleet histories, repair labels, and censoring [7]. Hydraulic Systems is a supplementary laboratory condition-monitoring demonstration with load-cycle component states [6]. SECOM is a supplementary high-dimensional production-quality demonstration with wafer-level yield labels [8].

This role assignment prevents a common mistake: row counts do not become independent failures. Table 2 records the empirical role, sampling structure, independent unit, independent positives, rows, and target type used for interpretation. The independent unit column is the primary evidential quantity for empirical interpretation.

Table 2: Dataset roles and evidential units.

| Dataset   | Role                   | Sampling             | Unit                                                       | Ind.<br>positives | Rows       | Target                                             |
|-----------|------------------------|----------------------|------------------------------------------------------------|-------------------|------------|----------------------------------------------------|
| Hydraulic | cycle<br>condition     | test-rig<br>cycles   | load cycle                                                 | 431               | 2,205      | condition<br>states                                |
| MetroPT   | event<br>warning       | ordered<br>telemetry | failure<br>episode /<br>operating<br>day, not<br>timestamp | 1                 | 10,773,588 | failure<br>labels                                  |
| MetroPT2  | event<br>warning       | ordered<br>telemetry | failure<br>episode /<br>operating<br>day, not<br>timestamp | 1                 | 7,116,940  | failure<br>labels                                  |
| SCANIA    | repair-risk<br>ranking | vehicle<br>histories | vehicle                                                    | 142               | 1,516,819  | published<br>labels and<br>time-to-<br>event files |
| SECOM     | yield failure          | wafer rows           | wafer /<br>production<br>example                           | 17                | 1,567      | yield labels                                       |

The supported estimands differ. MetroPT-style rows address event-level early warning on temporal telemetry splits. SCANIA addresses vehicle-level repair-risk ranking on a published test split. Hydraulic Systems addresses cycle-level component-condition detection. SECOM addresses production-example yield-failure detection. Hydraulic Systems and SECOM are therefore not interpreted as event-level replications of the compressor task. Broader predictive claims require coherent independent units and comparable targets, not only successful data preparation.

## 6 Experimental Protocol

The evaluation protocol is frozen before final experiment execution. It declares the train, validation, calibration, and test partitions; the exclusion and maintenance-boundary fields; warning horizons; alarm merge gap; matching tolerance; matching method; bootstrap unit; primary metrics; secondary metrics; and event utility costs.

Primary event metrics are event precision, event recall, event  $F_1$ , false-alarm events per operating day, duplicate alarms per failure, median warning lead time, and horizon Brier score. Secondary metrics include pointwise ranking metrics, time under warning, and event utility. Baselines are required to use the same causal feature availability and the same frozen event conversion, so a model is not rewarded for using future information or a more favorable matching rule.

Uncertainty is attached to the independent unit declared by the dataset and protocol. For simulation cells, the unit is the Monte Carlo repetition or registered replicate. For MetroPT-style case studies, the unit is the failure episode or operating interval, not the timestamp. For SCANIA, the relevant unit is the vehicle or repair history defined by the dataset release. For Hydraulic Systems, it is the load cycle with component-condition labels. For SECOM, it is the wafer or production example with a pass/fail yield outcome. Multiplicity is handled by keeping primary metrics and hypotheses predeclared and by reporting dataset-specific estimands.

The registered real-data matrix has three audit layers. The verification layer checks local raw and processed manifests, row counts, schema summaries, and declared independence units. The industrial-result layer writes one conservative diagnostic row per registered real dataset. The evidence-scope layer combines both outputs into a machine-readable summary of datasets, workflows, sampling structures, independent units, diagnostic statuses, and claim boundaries. This protocol is intentionally strict: a broader dataset base strengthens external coverage only when the estimand differences remain visible.

## 7 Results

### 7.1 Simulation validity and estimator stability

The first research question asks whether the extremal-index diagnostics are stable enough to interpret. The high-replication simulation artifact covers fourteen process families with 500 replications per configured cell, and a focused coverage artifact uses 1000 replications on a preregistered subset. The six estimators of Section 4 are compared against theoretical references where a closed form exists and numerical references otherwise, with unestablished configurations excluded from every performance quantity. Table 3 reports one representative process per clustering regime; the full grid and the reference-availability audit are in the supplement.

The central result is that the estimator ranking *inverts* between regimes. For the independent process, where  $\theta = 1$ , runs and K-gaps record interval coverage of 0.40 and 0.39 against a nominal 0.95, with biases of  $-0.133$  and  $-0.117$ . For the periodic target, where  $\theta = 1/2$ , the same two estimators are the least biased of the six, at  $-0.010$  and  $+0.003$ , with coverage of 0.844 and 0.850. A reader who saw only the near-independent rows would conclude against precisely the estimators that work once extremes actually cluster.

The apparent superiority of Ferro–Segers and disjoint blocks at  $\theta = 1$  requires care. Both attain coverage of 1.000 there, but with boundary-clipping probabilities between 0.50 and 0.70: their intervals contain the truth largely because the estimate saturates at the admissible boundary  $\theta = 1$ , not because they are calibrated. Figure 3 reports the clipping share beside the non-estimability share for this reason, and coverage at  $\theta = 1$  should not be read without it. The undercoverage of runs and K-gaps has the same origin: at the boundary the estimate can only err downwards, so a symmetric percentile interval misses systematically. Coverage comparisons

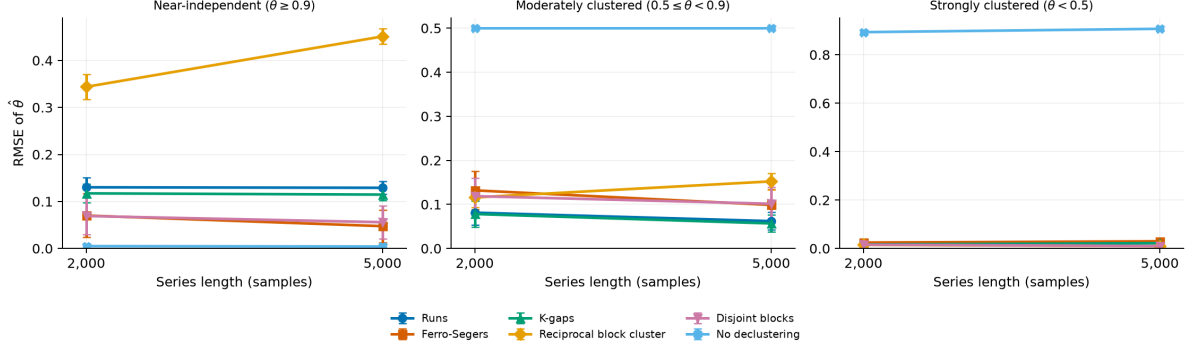


Figure 1: Finite-sample RMSE of the extremal-index estimate by series length, faceted by clustering regime. Error bars are 1.96 bootstrap standard errors. Panels use independent vertical scales because the no-declustering control has RMSE  $|1 - \theta|$  by construction. Reference values are  $\theta = 1$  (theoretical) for the near-independent panel,  $\theta = 1/2$  (theoretical) for the moderately clustered panel, and numerically established values below 0.3 for the strongly clustered panel.

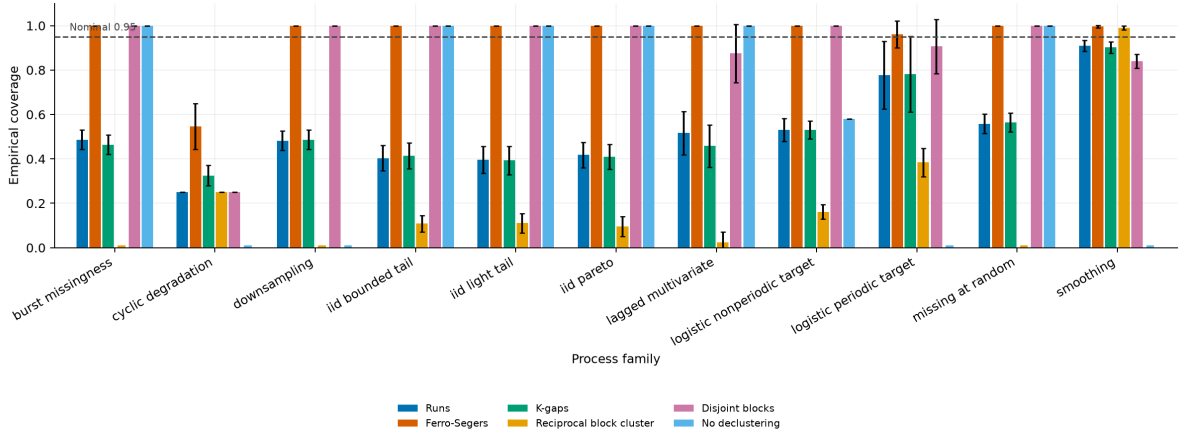


Figure 2: Empirical coverage of the bootstrap interval by process family against the nominal 0.95 level, with 1.96 binomial standard errors. Families whose reference extremal index could not be established are omitted, since coverage is undefined without a reference.

at  $\theta = 1$  are structurally distorted for every estimator, which is a further reason to treat the near-independent grid as bounded evidence.

The no-declustering control behaves exactly as intended, with zero bias at  $\theta = 1$  and bias  $+0.500$  and  $+0.728$  in the two clustered rows, confirming that the comparison is measuring extremal clustering rather than an artifact of the harness.

Threshold and run-length sensitivity in Figure 4 and Table 4 show that clustering conclusions depend on  $q$  and  $m$ . That supports a negative-result interpretation: regime-conditioned dynamical-EVT diagnostics are sensitive design objects, not model-free evidence of impending failure.

## 7.2 Alarm conversion and failed-detection regions

The second research question asks whether declustering and alarm merging reduce duplicate alarms while preserving event recall. In the registered threshold grid, event recall was zero at all reported quantiles, with false-alarm burdens between 133.92 and 168.48 alarm episodes per operating day. This is a failed-detection region under the current observable, threshold grid, and matching rule; it shows that alarm conversion can destroy utility. The full threshold grid

Table 3: Finite-sample extremal-index estimator performance for one representative process per clustering regime, aggregated over the threshold span shown. Coverage is measured against a nominal 0.95 level; the failure column is the share of replicates returning no estimate. A dagger marks a numerically established reference extremal index; unmarked references are closed-form. The full grid is in the supplement.

| Estimator                                                                 | $n$  | Threshold | Bias   | RMSE  | Coverage | Fail. |
|---------------------------------------------------------------------------|------|-----------|--------|-------|----------|-------|
| <i>iid light tail, <math>\theta_{\text{ref}} = 1.000</math></i>           |      |           |        |       |          |       |
| Runs                                                                      | 5000 | 0.95-0.99 | -0.133 | 0.139 | 0.397    | 0.000 |
| Ferro-Segers                                                              | 5000 | 0.95-0.99 | -0.025 | 0.051 | 1.000    | 0.000 |
| K-gaps                                                                    | 5000 | 0.95-0.99 | -0.117 | 0.123 | 0.393    | 0.000 |
| Reciprocal block cluster                                                  | 5000 | 0.95-0.99 | -0.423 | 0.427 | 0.111    | 0.000 |
| Disjoint blocks                                                           | 5000 | 0.95-0.99 | -0.030 | 0.058 | 1.000    | 0.000 |
| No declustering                                                           | 5000 | 0.95-0.99 | 0.000  | 0.000 | 1.000    | 0.000 |
| <i>logistic periodic target, <math>\theta_{\text{ref}} = 0.500</math></i> |      |           |        |       |          |       |
| Runs                                                                      | 5000 | 0.95-0.99 | -0.010 | 0.072 | 0.844    | 0.000 |
| Ferro-Segers                                                              | 5000 | 0.95-0.99 | 0.064  | 0.115 | 0.933    | 0.000 |
| K-gaps                                                                    | 5000 | 0.95-0.99 | 0.003  | 0.067 | 0.850    | 0.000 |
| Reciprocal block cluster                                                  | 5000 | 0.95-0.99 | -0.113 | 0.134 | 0.578    | 0.000 |
| Disjoint blocks                                                           | 5000 | 0.95-0.99 | 0.065  | 0.110 | 0.839    | 0.000 |
| No declustering                                                           | 5000 | 0.95-0.99 | 0.500  | 0.500 | 0.000    | 0.000 |
| <i>smoothing, <math>\theta_{\text{ref}} = 0.272^\dagger</math></i>        |      |           |        |       |          |       |
| Runs                                                                      | 2000 | 0.98      | -0.012 | 0.036 | 0.910    | 0.000 |
| Ferro-Segers                                                              | 2000 | 0.98      | 0.019  | 0.076 | 0.996    | 0.000 |
| K-gaps                                                                    | 2000 | 0.98      | -0.009 | 0.039 | 0.902    | 0.000 |
| Reciprocal block cluster                                                  | 2000 | 0.98      | -0.015 | 0.041 | 0.990    | 0.000 |
| Disjoint blocks                                                           | 2000 | 0.98      | 0.021  | 0.053 | 0.840    | 0.000 |
| No declustering                                                           | 2000 | 0.98      | 0.728  | 0.728 | 0.000    | 0.000 |

Table 4: Compact threshold and run-length stability summary.

| q range  | run range | theta range | cluster-count range | conclusion                            |
|----------|-----------|-------------|---------------------|---------------------------------------|
| 0.9–0.99 | 0–20      | 0.019–0.64  | 31–157              | sensitive to threshold and run length |

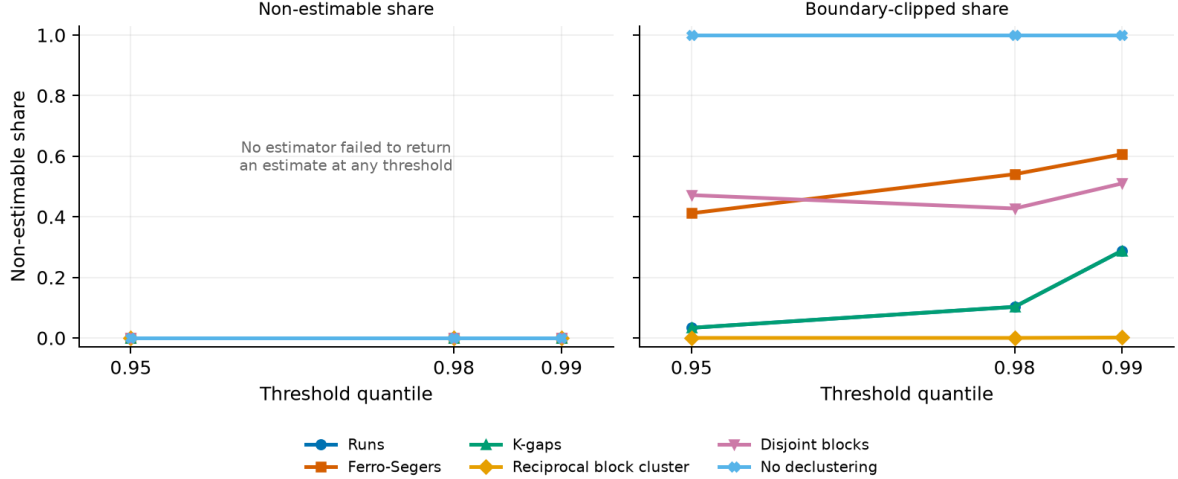


Figure 3: Non-estimability and boundary clipping against threshold quantile. No estimator failed to return a value at any threshold; the informative panel is the clipped share, which explains the coverage of the block-based estimators near  $\theta = 1$ .

is reported in the supplement rather than as a main-text figure because a zero-recall line plot would add no scientific information.

Across all registered combinations of alarm merge gap and post-onset matching tolerance, event recall, precision, and  $F_1$  remained zero. Relaxing post-onset tolerance did not alter the result, so the full grid is reported in the supplement rather than as a main-text sensitivity figure.

Table 5: Invariant failed-detection summary across alarm-conversion settings.

| merge-gap<br>range | tolerance<br>range | combinations | recall range | precision<br>range | F1 range |
|--------------------|--------------------|--------------|--------------|--------------------|----------|
| 0-60               | 0-300              | 20           | 0            | 0                  | 0        |

### 7.3 Held-out event-level method comparison

The third and fourth research questions ask whether the registered robust extreme-score diagnostic adds operational value after episode conversion, and why empirical failures occur. Table 7 gives the held-out event-level rows for MetroPT and MetroPT2. Each dataset contains one labelled test event in the registered split. Both events are detected by the registered score, but the detection is operationally weak: MetroPT produces 2631 predicted alarm episodes with precision 0.00038, and MetroPT2 produces 852 predicted alarm episodes with precision 0.00117. The median lead times are not persuasive by themselves because they are paired with extreme false-alarm burdens.

### 7.4 Complete held-out event-level benchmark

Nineteen methods were run on both compressor datasets under one shared alarm policy: identical warning horizon, matching tolerance, matching rule, operating exposure and independent unit, verified by the fairness audit in Supplementary Table S4. Alarm merging is treated as part of the detector rather than the protocol, since four methods exist precisely to vary declustering; the audit reports which event conversions were used instead of requiring one. Supplementary Table S2 reports the complete held-out event-level benchmark under the shared alarm policy,

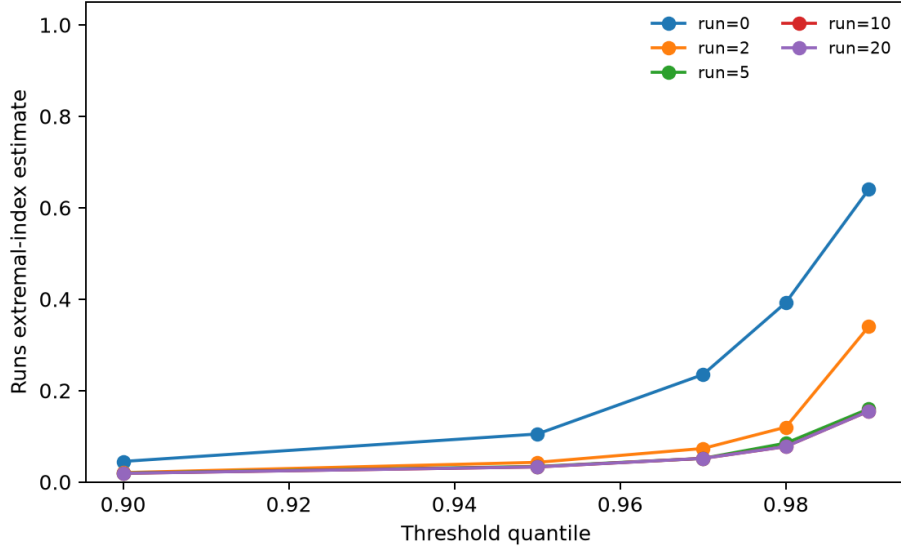


Figure 4: Threshold and run-length stability diagnostic for the registered recurrence observable.

and Supplementary Table S3 records each method’s score, threshold rule, event policy, merge gap, parameter count, calibration status and tuning partition.

Table 6: Compact held-out event-level benchmark: the best method in each category per dataset, selected on event  $F_1$  with alarm burden as the declared tie-break. A method flagged as effectively always on cannot win a category, because episode-level precision and recall are degenerate for a permanently-on detector. The complete benchmark over all nineteen methods, including any excluded on that ground, is in Supplementary Table S2.

| Category          | Method                   | Det. | Precision | FA/day | Lead  | Utility |
|-------------------|--------------------------|------|-----------|--------|-------|---------|
| <i>MetroPT</i>    |                          |      |           |        |       |         |
| Simple            | Single-sensor threshold  | yes  | 0.000441  | 90.9   | 2,333 | -452    |
| Classical EVT     | Classical POT/GPD        | yes  | 0.000396  | 101    | 3,336 | -503    |
| Anomaly           | Conformal anomaly        | yes  | 0.000396  | 101    | 3,346 | -504    |
| Learned           | Linear autoencoder       | yes  | 0.000212  | 189    | 3,433 | -941    |
| Registered        | Recurrence score         | yes  | 0.00038   | 105    | 3,346 | -525    |
| Prototype region  | Failure-prototype region | no   | 0         | 0      |       | -2      |
| Rare-state region | Rare-state region        | yes  | 0.000396  | 101    | 3,346 | -504    |
| <i>MetroPT2</i>   |                          |      |           |        |       |         |
| Simple            | Single-sensor threshold  | yes  | 0.00282   | 21.2   | 3,444 | -69.3   |
| Classical EVT     | Classical POT/GPD        | yes  | 0.00119   | 50.6   | 3,457 | -166    |
| Anomaly           | Conformal anomaly        | yes  | 0.00118   | 51.3   | 3,469 | -168    |
| Learned           | Nonlinear autoencoder    | yes  | 0.00115   | 52.6   | 3,457 | -173    |
| Registered        | Recurrence score         | yes  | 0.00117   | 51.5   | 3,469 | -169    |
| Prototype region  | Failure-prototype region | yes  | 0.00637   | 9.41   | 3,435 | -30.1   |
| Rare-state region | Rare-state region        | yes  | 0.00118   | 51.4   | 3,469 | -169    |

Five conclusions follow, and none of them is favourable to the registered method.

*No method is operationally plausible on either dataset.* Every method that detects the held-out failure does so with event precision below 0.7%, and most below 0.05%. The best event utility achieved by any detecting method is negative on both datasets. A maintenance function receiving these alarms would be responding overwhelmingly to false positives.

*The best simple baseline wins on both datasets.* On MetroPT the best simple policy is the single-sensor threshold, at 90.9 false alarms per operating day against the registered score’s 105.3; the fixed engineering threshold reaches a lower false-alarm rate of 62.0 but is flagged as

effectively always on, holding the system under warning for 75.1% of the test period, so it is excluded from selection. On MetroPT2 the single-sensor threshold is best again and by a wide margin, at 21.2 false alarms per day against 51.5 for the registered score, with more than double its event precision.

*Classical EVT and anomaly baselines do not change the picture.* The generalised Pareto return level, the three declustering policies, SPOT, the isolation forest, the change-point detector, both autoencoders and the conformal detector all sit within the same precision decade as the registered score. Making the tail model or the declustering rule more sophisticated does not recover operational value that the alarm conversion has already destroyed.

*The registered recurrence score adds no measurable event-level value.* It is beaten by a simple single-sensor threshold on both datasets, on precision, on false alarms per day and on event utility. Its distinguishing feature is not accuracy but exposure: at 0.12% and 0.06% of the test period under warning it is far less intrusive than the engineering threshold, which is a real property but not the early-warning claim the method was proposed to support.

*The one exception is the failure-prototype region on MetroPT2, and it does not transfer.* There it attains the best precision of any method, 0.0064, at 9.4 false alarms per day and the best utility,  $-30.1$ . On MetroPT the same construction raises no alarms at all and misses the failure entirely. A method that is the strongest performer on one compressor dataset and non-detecting on the other cannot be reported as a transferable result.

Alarm burden must be read on two axes that can disagree. Episode counts measure how often an operator is interrupted; time under warning measures how long the system is held in an alarmed state. On MetroPT the engineering threshold raises fewer episodes than the registered score (1551 against 2631) while holding the system under warning for 75.1% of the test period against the registered score’s 0.12%. A detector that merges its exceedances into few long alarms therefore looks better on episode counts and far worse on exposure. Because episode-level precision and recall are computed after merging, a permanently-on detector attains a recall and precision of one with no unmatched episodes to count as false alarms; exposure is the only reported quantity that exposes it. Every alarm-burden comparison in this paper is therefore reported with both axes, and the supplementary benchmark flags any method whose warning exposure exceeds half the test period.

Table 8 compares the registered score with a compact event-level baseline set under the same held-out event policy. The table keeps the best simple rule class, a classical EVT baseline, an anomaly baseline, the registered recurrence score, and the two target-region methods. The full baseline family remains in the generated CSV artifact.

Table 7: Held-out event-level diagnostic rows for compressor datasets.

| Dataset  | Failure                                        | Target | Alarms | Recall | Precision | FA/day | Exposure<br>% | Lead  |
|----------|------------------------------------------------|--------|--------|--------|-----------|--------|---------------|-------|
| MetroPT  | MetroPT<br>regis-<br>tered<br>test<br>failure  | 1      | 2,631  | 1      | 0.00038   | 105    | 0.12          | 3,346 |
| MetroPT2 | MetroPT2<br>regis-<br>tered<br>test<br>failure | 1      | 852    | 1      | 0.00117   | 51.5   | 0.06          | 3,469 |

Table 8: Real held-out event-level comparison under the shared alarm policy.

| Data     | Method                        | Family           | Target | Alarms | Recall | Precision | FA/day | Lead               |
|----------|-------------------------------|------------------|--------|--------|--------|-----------|--------|--------------------|
| MetroPT  | Classical<br>POT/GPD          | baseline         | 1      | 2,523  | 1      | 0.000396  | 101    | 3,336              |
| MetroPT  | Engineering<br>thresh-<br>old | baseline         | 1      | 1,551  | 1      | 0.000645  | 62     | 3,499              |
| MetroPT  | Isolation<br>Forest           | baseline         | 1      | 2,577  | 1      | 0.000388  | 103    | 3,432              |
| MetroPT  | Dynamical<br>EVT<br>score     | registered       | 1      | 2,631  | 1      | 0.00038   | 105    | 3,346              |
| MetroPT  | Failure<br>proto-<br>type     | target<br>region | 1      | 0      | 0      | 0         | 0      | Not es-<br>timable |
| MetroPT  | Rare-<br>state<br>region      | target<br>region | 1      | 2,527  | 1      | 0.000396  | 101    | 3,346              |
| MetroPT2 | Classical<br>POT/GPD          | baseline         | 1      | 837    | 1      | 0.00119   | 50.6   | 3,457              |
| MetroPT2 | Engineering<br>thresh-<br>old | baseline         | 1      | 614    | 1      | 0.00163   | 37.2   | 7,264              |
| MetroPT2 | Isolation<br>Forest           | baseline         | 1      | 1,742  | 1      | 0.000574  | 105    | 3,022              |
| MetroPT2 | Dynamical<br>EVT<br>score     | registered       | 1      | 852    | 1      | 0.00117   | 51.5   | 3,469              |
| MetroPT2 | Failure<br>proto-<br>type     | target<br>region | 1      | 157    | 1      | 0.00637   | 9.41   | 3,435              |
| MetroPT2 | Rare-<br>state<br>region      | target<br>region | 1      | 851    | 1      | 0.00118   | 51.4   | 3,469              |

## 7.5 Target-region heterogeneity across MetroPT and MetroPT2

The strongest empirical divergence is the failure-prototype target region. It materially reduces alarm burden in MetroPT2 but fails to detect the held-out MetroPT event. Table 9 reports the within-dataset and cross-dataset target-region comparison for failure-prototype and rare-state regions. This instability means the two compressor datasets cannot be treated as equivalent evidence for one transferable recurrence region.

Table 9: Target-region behaviour across MetroPT and MetroPT2 held-out event splits.

| Source   | Target   | Region            | Recall | Precision | FA/day | Transfer                        |
|----------|----------|-------------------|--------|-----------|--------|---------------------------------|
| MetroPT  | MetroPT  | Failure prototype | 0      | 0         | 0      | within_dataset                  |
| MetroPT  | MetroPT  | Rare-state region | 1      | 0.000396  | 101    | within_dataset                  |
| MetroPT  | MetroPT2 | Failure prototype | 1      | 0.00637   | 9.41   | cross_dataset_metric_projection |
| MetroPT  | MetroPT2 | Rare-state region | 1      | 0.00118   | 51.4   | cross_dataset_metric_projection |
| MetroPT2 | MetroPT  | Failure prototype | 0      | 0         | 0      | cross_dataset_metric_projection |
| MetroPT2 | MetroPT  | Rare-state region | 1      | 0.000396  | 101    | cross_dataset_metric_projection |
| MetroPT2 | MetroPT2 | Failure prototype | 1      | 0.00637   | 9.41   | within_dataset                  |
| MetroPT2 | MetroPT2 | Rare-state region | 1      | 0.00118   | 51.4   | within_dataset                  |

## 7.6 Matched negative controls

Table 10 compares the observed failure-prototype alarm burden with 500 deterministic matched-control draws for each control family and dataset. These controls preserve occupancy, exposure length, temporal shifting, rare-region structure, or block structure while breaking the specific relationship to the held-out failure. The comparison is descriptive because only two independent held-out compressor failures are available.

## 7.7 Score-threshold-alarm decomposition

Figure 5 replaces the previous incomparable metric plot. It follows count-valued evidence through the score, threshold, alarm, and target-event layers. The dominant failure layer is not identical across methods: some scores never isolate the event, some thresholds create too many exceedances, and some alarm policies collapse precision after clustering and merging.

## 7.8 Timeline reconciliation

The event timeline in Figure 6 is generated from the MetroPT held-out test failure. It is a local explanation of one failure window, not a substitute for global false-alarm burden. The global alarm counts remain those in the event-level tables.

## 7.9 Supplementary non-event stress tests

Hydraulic Systems, SCANIA Component X, and SECOM are reported separately in Table 11 because their independent units are cycles, vehicles, and wafers rather than compressor failure episodes. All three have  $F_1 = 0$  under the conservative high-quantile diagnostic. These are not

Table 10: Observed failure-prototype alarm burden against 500 matched-control draws per family.

| Data     | Observed             | Control                       | Observed<br>FA/day | Control<br>median | 95%<br>interval | Percentile | Interpretation                               |
|----------|----------------------|-------------------------------|--------------------|-------------------|-----------------|------------|----------------------------------------------|
| MetroPT  | Failure<br>prototype | Label<br>permuta-<br>tion     | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT  | Failure<br>prototype | Phase-<br>randomised          | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT  | Failure<br>prototype | Prototype<br>permuta-<br>tion | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT  | Failure<br>prototype | Random<br>occu-<br>pancy      | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT  | Failure<br>prototype | Rare-<br>region<br>matched    | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT  | Failure<br>prototype | Time-<br>shifted              | 0                  | 0                 | 0-0             | 1          | not lower<br>burden                          |
| MetroPT2 | Failure<br>prototype | Label<br>permuta-<br>tion     | 9.41               | 9.53              | 9.35-9.59       | 0.142      | lower<br>alarm<br>burden<br>than<br>controls |
| MetroPT2 | Failure<br>prototype | Phase-<br>randomised          | 9.41               | 51.5              | 51.3-51.6       | 0          | lower<br>alarm<br>burden<br>than<br>controls |
| MetroPT2 | Failure<br>prototype | Prototype<br>permuta-<br>tion | 9.41               | 8.01              | 7.53-8.44       | 1          | not lower<br>burden                          |
| MetroPT2 | Failure<br>prototype | Random<br>occu-<br>pancy      | 9.41               | 8.01              | 7.41-8.5        | 1          | not lower<br>burden                          |
| MetroPT2 | Failure<br>prototype | Rare-<br>region<br>matched    | 9.41               | 50.5              | 49.9-51         | 0          | lower<br>alarm<br>burden<br>than<br>controls |
| MetroPT2 | Failure<br>prototype | Time-<br>shifted              | 9.41               | 9.53              | 9.35-9.59       | 0.142      | lower<br>alarm<br>burden<br>than<br>controls |

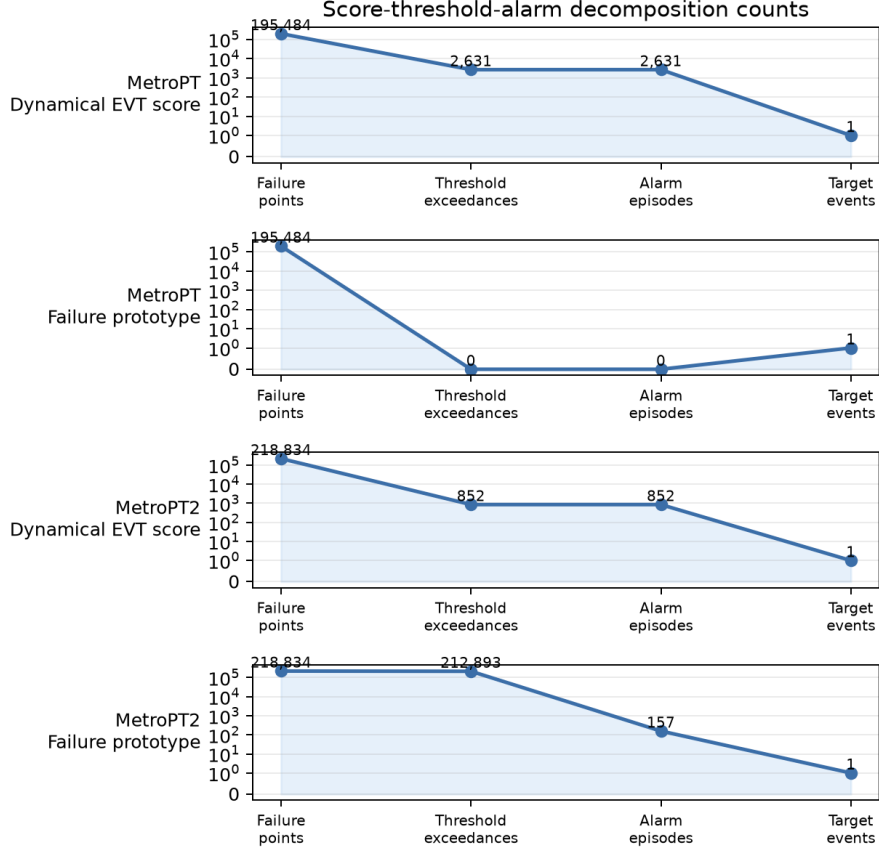


Figure 5: Score-threshold-alarm decomposition for real held-out event methods. The panels show count-valued transitions rather than mixing event precision with false alarms per day on one axis.

hidden failures. They are evidence that a simple extreme-score adapter does not transfer across laboratory cycle states, vehicle-level repair risk, and wafer-level yield labels without changing the estimand and evaluation design. Table 12 summarizes the most plausible explanations under the current evidence.

Table 11: Non-event real-data diagnostic rows kept separate from event alarms.

| Data           | Unit    | Units | Target | Pred | Recall | Precision | F1 | Brier  | Limit              |
|----------------|---------|-------|--------|------|--------|-----------|----|--------|--------------------|
| Hydraulic load | cycle   | 441   | 431    | 0    | 0      | 0         | 0  | 0.977  | cycle labels       |
| SCANIA         | vehicle | 5,045 | 142    | 132  | 0      | 0         | 0  | 0.0543 | estimand mis-match |
| SECOM          | wafer   | 314   | 17     | 10   | 0      | 0         | 0  | 0.086  | yield target       |

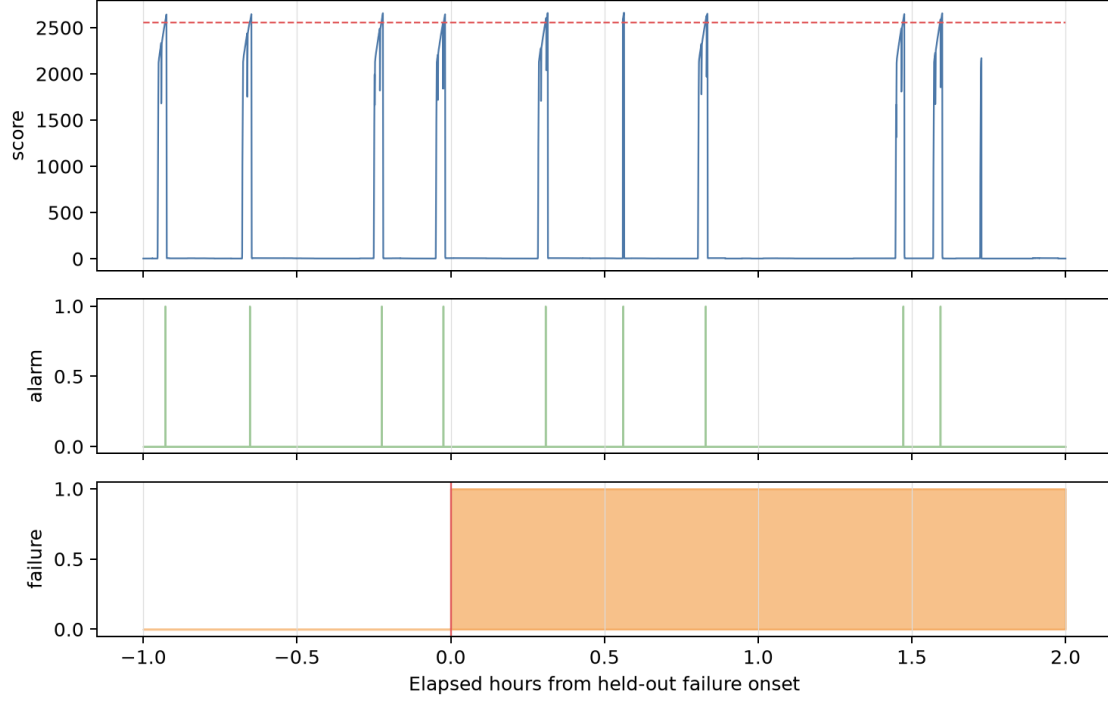


Figure 6: Traceable MetroPT held-out failure timeline. The x-axis is elapsed hours from the registered test failure onset; the panels show the fitted score threshold, alarm flags after thresholding, and labelled failure interval.

Table 12: Most plausible explanations under the current evidence.

| Dataset   | Primary explanation                                                 | Evidence                                     | Evidence status           | Competing explanation                           | Confidence |
|-----------|---------------------------------------------------------------------|----------------------------------------------|---------------------------|-------------------------------------------------|------------|
| Hydraulic | cycle-state target is not a field event onset                       | 431 positives, F1 0                          | diagnostic evidence       | component state needs supervised feature family | low        |
| MetroPT   | alarm-conversion burden with one held-out failure                   | recall 1.00, precision 0.00038, 2,631 alarms | direct event-row evidence | threshold too low or observable too broad       | medium     |
| MetroPT2  | alarm-conversion burden with one held-out failure                   | recall 1.00, precision 0.00117, 852 alarms   | direct event-row evidence | threshold too low or observable too broad       | medium     |
| SCANIA    | vehicle-level repair-risk estimand mismatch                         | 5,045 vehicles, F1 0                         | estimand evidence         | features insufficient for repair ranking        | low        |
| SECOM     | yield-failure target and missingness differ from maintenance events | 17 positives, F1 0                           | diagnostic evidence       | missingness and class imbalance dominate        | low        |

## 8 Discussion

### 8.1 Estimator finite-sample validity

The Monte Carlo results support a bounded interpretation of the EVT diagnostics. Runs, Ferro-Segers, K-gaps, reciprocal mean cluster size, block clustering, and no-declustering references are useful for detecting estimator instability across process families, thresholds, missingness, and dynamical structure. They do not by themselves identify maintenance-relevant degradation. The practical implication is that recurrence estimates should be treated as conditional diagnostics tied to the chosen observable, threshold, run length, and sampling process.

### 8.2 Target-region transferability

The central empirical finding is target-region heterogeneity. The failure-prototype region missed the held-out MetroPT failure, yet detected the held-out MetroPT2 failure with 157 alarm episodes and 9.41 false alarms per operating day. That result is materially better than the registered recurrence score on MetroPT2, but it does not transfer back to MetroPT. The evidence therefore supports target-region instability, not a general target-region solution.

### 8.3 Score, Threshold, and Alarm Failure

The failure mode is layered. Some methods fail at the score layer because failure-proximal states are not separated from normal states. Others produce score information that is lost or made operationally unusable by threshold choice. The registered recurrence score detects both compressor failures, but alarm conversion leaves hundreds or thousands of episodes. That is an alarm-policy failure after thresholding, not a calibrated maintenance signal.

### 8.4 Comparison With Baselines

Simple baselines remain competitive under the held-out event policy. The engineering threshold detected both compressor events, and its burden relative to the registered score depends on the axis measured: fewer alarm episodes, but far greater warning exposure. That asymmetry is itself the operational point. A method can be made to look better on episode counts by merging its alarms into fewer, longer blocks, which is not an improvement an operator would recognise, so claiming a burden advantage requires naming the burden. Classical EVT and anomaly baselines did not remove the basic precision problem. The proposed recurrence score therefore adds diagnostic structure, but it does not add measurable operational value on these held-out events.

### 8.5 Matched Negative Controls

Matched controls prevent over-interpreting one favorable target-region result. For MetroPT2, the failure-prototype region has lower false-alarm burden than rare-region and phase-randomised controls, but occupancy-matched random regions can have lower burden. The result is promising as a diagnostic contrast, not as evidence that the learned region is uniquely failure-proximal.

### 8.6 Independent-Event Scarcity

Only two independent held-out compressor failures are available. This prevents stable estimation of event-level uncertainty, calibration, and failure-type heterogeneity. Timestamp counts cannot substitute for independent failures because alarm episodes are temporally dependent and can cluster around the same operating regime.

## 8.7 Operational Implications

The workflow is useful as an audit of how extreme-score methods fail. It exposes whether apparent score excursions survive thresholding, declustering, alarm merging, and event matching. It does not establish deployable early warning. Deployment would require prospectively validated target regions, enough independent failures or entities, acceptable false-alarm burden, useful lead time, and comparison against simple policies under the same episode-level evaluation.

## 9 Conclusion

This study demonstrated a reproducible way to test whether clustered extremes become early warnings, and the answer on the evaluated public data is mostly negative. The registered recurrence score detected the two held-out compressor failures, but only with 2631 and 852 alarm episodes and false-alarm burdens of 105.34 and 51.47 alarms per operating day. These detections are not operationally plausible maintenance signals. Nor is the comparison against simple policies settled by episode counts alone: the engineering threshold raises fewer episodes while keeping the system under warning for most of the MetroPT test period, so alarm burden is only interpretable when interruption frequency and warning exposure are reported together.

The target-region evidence is the most informative positive contrast. The failure-prototype region missed the held-out MetroPT event, but detected the held-out MetroPT2 event with 157 alarms and 9.41 false alarms per day. That divergence shows that recurrence regions can matter, but it also shows that their failure-proximal meaning is not stable across the two compressor datasets. Matched controls further narrow the interpretation: the MetroPT2 prototype result is better than several structure-preserving controls, but not uniformly better than occupancy-matched random regions.

The main failure mechanisms are now separated. Score quality, threshold selection, and alarm conversion can each destroy utility. In the registered compressor score, threshold exceedances can recover the event, but alarm conversion collapses precision. In the failure-prototype region, MetroPT fails earlier: the held-out failure does not enter the learned region. Non-event datasets add stress-test evidence only; their cycles, vehicles, and wafers are not independent compressor failure episodes.

Target-region methods add diagnostic value by exposing heterogeneity between MetroPT and MetroPT2, but no evaluated method establishes reliable operational early warning. The evidence required next is specific: more independent failure episodes, prospectively frozen target regions, stable regime definitions, deployment-realistic alarm policies, and external validation showing acceptable recall, false-alarm burden, lead time, and utility under the same event-level protocol.

## Code and Data Availability

Code, configuration files, data-acquisition scripts, generated-paper tooling, and reproducibility materials are maintained in the project repository [9]. Raw third-party datasets are not redistributed by this repository; the acquisition pipeline stores them locally under `data/raw/` and documents their public sources through the dataset registry.

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